

Time Series Model Identification — Exploratory Case Studies

In this session, we will approach time-series modelling as an **exploratory investigation**.

Instead of being told which model to use, we will receive a time series and investigate:

1. Is the series stationary?
2. If it is not stationary, should we difference it?
3. What do the ACF and PACF suggest?
4. Which model orders are plausible?
5. Which candidate model has better AIC/BIC?
6. Do the residuals look like white noise?
7. Is the final model good?

The datasets in this notebook are simulated from known processes, but the generating model is hidden while we investigate each case.

Main idea: ACF/PACF help us form hypotheses. AIC/BIC help us compare candidates. Residual diagnostics tell us whether the model has captured the predictable structure.

1. Models We Need to Recognize

AR(p) — Autoregressive Model

An AR(p) model explains the current value using previous observations:

$$X_t = c + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \cdots + \phi_p X_{t-p} + \epsilon_t$$

Typical identification pattern

- **PACF cuts off** around lag p
- ACF tends to **tail off**

For example, if the PACF has a clear cutoff after lag 3, AR(3) is a natural candidate.

MA(q) — Moving Average Model

An MA(q) model explains the current value using current and previous shocks:

$$X_t = \mu + \epsilon_t + \theta_1 \epsilon_{t-1} + \dots + \theta_q \epsilon_{t-q}$$

Typical identification pattern

- **ACF cuts off** around lag q
- PACF tends to **tail off**

For example, if the ACF cuts off after lag 4, MA(4) is a natural candidate.

ARMA(p,q)

ARMA combines autoregressive and moving-average components.

For an ARMA model:

- ACF generally tails off
- PACF generally tails off

There is usually no clean cutoff that uniquely identifies both orders, so we generate a small set of plausible candidates and compare them.

ARIMA(p,d,q)

ARIMA adds differencing:

$$\phi(B)(1 - B)^d X_t = \theta(B)\epsilon_t$$

where:

- p = AR order
- d = number of differences
- q = MA order

The important idea is:

Determine d first, then identify the ARMA structure of the differenced series.

For example:

$$ARIMA(1, 1, 2)$$

means:

1. Difference the original series once.
2. Model the differenced series using ARMA(1,2).

2. The Model Identification Workflow

For every case study, we will follow the same process.

Step 1 — Plot the original series

Ask:

- Is there a trend?
 - Does the mean appear constant?
 - Is the variance roughly constant?
 - Is there seasonality?
 - Are there unusual observations?
-

Step 2 — Check stationarity

We will use:

- visual inspection
- the Augmented Dickey-Fuller (ADF) test

The ADF test has a unit-root null hypothesis.

- Small p-value → evidence against a unit root
- Large p-value → insufficient evidence to reject a unit root

The test should support our reasoning rather than replace it.

Step 3 — Difference if necessary

If the series is non-stationary because of a stochastic trend, we may difference it.

Then we check the differenced series again.

Step 4 — Inspect ACF and PACF

For a stationary series:

Model	ACF	PACF
AR(p)	Tails off	Cuts off around p
MA(q)	Cuts off around q	Tails off

Model	ACF	PACF
ARMA(p,q)	Tails off	Tails off

These are **rules of thumb**, not mathematical guarantees.

Step 5 — Compare candidate models

We fit a small number of plausible models and compare:

- AIC
- BIC

Lower values are generally preferred.

Step 6 — Diagnose the residuals

A good model should leave residuals that are approximately **white noise**.

We will check:

- residual time plot
- residual distribution
- residual ACF
- Ljung-Box test

If substantial autocorrelation remains in the residuals, the model has probably missed some predictable structure.

Overall workflow

Stationarity → **Differencing** → **ACF/PACF** → **Candidate models** → **AIC/BIC** → **Residual diagnostics**

AIC and BIC — Comparing Candidate Models

After examining the ACF and PACF, we will usually have **several plausible models**.

For example, suppose our analysis suggests that the series could be:

- AR(1)
- AR(2)



- AR(3)
- ARMA(1,1)

How do we decide which one to use?

This is where **AIC** and **BIC** are useful.

The basic idea

A more complicated model will usually fit the observed data better because it has more parameters.

For example:

$$AR(1) \rightarrow 1 \text{ AR parameter}$$

while

$$AR(5) \rightarrow 5 \text{ AR parameters}$$

The AR(5) model has more flexibility, so it may achieve a better fit simply because it is more complicated.

This can lead to **overfitting**.

AIC and BIC address this by balancing two things:

■ **How well does the model fit the data?**

and

■ **How complicated is the model?**

In other words:

$$\text{Good model} = \text{Good fit} + \text{Reasonable complexity}$$

AIC — Akaike Information Criterion

AIC is defined as:

$$AIC = 2k - 2\ln(L)$$

where:

- k = number of estimated parameters
- L = likelihood of the model

The first term penalizes the model for having more parameters, while the likelihood term rewards better fit.

Lower AIC is better.

BIC — Bayesian Information Criterion

BIC is defined as:

$$BIC = k \ln(n) - 2 \ln(L)$$

where:

- k = number of estimated parameters
- n = number of observations
- L = likelihood of the model

BIC penalizes additional parameters more strongly than AIC, especially when the sample size is large.

Lower BIC is better.

AIC vs BIC

	AIC	BIC
Rewards better fit	✓	✓
Penalizes complexity	✓	✓
Complexity penalty	Smaller	Larger
Tends to prefer	More flexible models	More parsimonious models

Therefore, it is possible for AIC and BIC to select different models.

Example

Suppose we fit four candidate models:

Model	AIC	BIC
AR(1)	1420	1430
AR(2)	1410	1425

Model	AIC	BIC
AR(3)	1412	1431
ARMA(2,1)	1408	1432

For AIC:

$$1408 < 1410$$

so **ARMA(2,1)** has the lowest AIC.

For BIC:

$$1425 < 1430 < 1431 < 1432$$

so **AR(2)** has the lowest BIC.

This happens because BIC applies a stronger penalty to the additional parameters in the more complicated models.

Important: AIC/BIC do NOT tell us whether a model is good

AIC and BIC are **relative measures**.

We should not say:

"An AIC of 100 is good."

Instead, we say:

"Among the models we compared on the same dataset, this model has the lowest AIC."

Therefore, our complete model-selection process is:

ACF/PACF → Candidate Models → AIC/BIC → Residual Diagnostics

ACF and PACF help us **generate candidates**.

AIC and BIC help us **compare candidates**.

Residual diagnostics help us determine whether the selected model is **adequate**.

Key takeaway

Lower AIC/BIC is better, but the model with the lowest AIC/BIC is not

```
In [1]: import warnings
warnings.filterwarnings("ignore")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from statsmodels.tsa.arima_process import ArmaProcess
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.tsa.stattools import adfuller
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
from statsmodels.stats.diagnostic import acorr_ljungbox

np.random.seed(42)
```

3. Helper Functions

We will create a few functions so that we can apply the same workflow to every dataset.

The important part of the session is **not memorizing these functions**.

The important part is understanding the reasoning behind the workflow.

```
In [9]: def plot_series(x, title):
    plt.figure(figsize=(12, 4))
    plt.plot(x)
    plt.title(title)
    plt.xlabel("Time")
    plt.ylabel("Value")
    plt.show()

def adf_report(x, name="Series"):
    result = adfuller(x.dropna())

    print(f"{name}")
    print(f"ADF statistic: {result[0]:.4f}")
    print(f"p-value: {result[1]:.4f}")

    if result[1] < 0.05:
        print("Conclusion: evidence against a unit root → stationary.")
    else:
        print("Conclusion: insufficient evidence to reject the unit-root null → non")

def plot_acf_pacf(x, title, lags=30):
    fig, axes = plt.subplots(1, 2, figsize=(14, 4))

    plot_acf(
        x.dropna(),
        lags=lags,
        ax=axes[0]
```



```

)

plot_pacf(
    x.dropna(),
    lags=lags,
    ax=axes[1],
    method="ywm"
)

axes[0].set_title(f"{title} - ACF")
axes[1].set_title(f"{title} - PACF")

plt.tight_layout()
plt.show()

```

```

In [10]: def compare_models(x, orders):
        """
        Fit a collection of ARIMA models and compare AIC/BIC.
        """

        results = []

        for order in orders:

            try:
                fitted = ARIMA(
                    x,
                    order=order
                ).fit()

                results.append({
                    "Model": f"ARIMA{order}",
                    "p": order[0],
                    "d": order[1],
                    "q": order[2],
                    "AIC": fitted.aic,
                    "BIC": fitted.bic
                })

            except Exception:
                pass

        results = pd.DataFrame(results)

        return results.sort_values(
            "AIC"
        ).reset_index(drop=True)

```

```

In [11]: def residual_diagnostics(fitted):

        residuals = fitted.resid.dropna()

        fig, axes = plt.subplots(
            1, 3,
            figsize=(16, 4)

```

```

)

# Residual time series
axes[0].plot(residuals)
axes[0].axhline(
    0,
    linestyle="--"
)
axes[0].set_title("Residuals")

# Residual distribution
axes[1].hist(
    residuals,
    bins=30
)
axes[1].set_title("Residual Distribution")

# Residual ACF
plot_acf(
    residuals,
    lags=24,
    ax=axes[2]
)
axes[2].set_title("Residual ACF")

plt.tight_layout()
plt.show()

print("Ljung-Box Test")

lb = acorr_ljungbox(
    residuals,
    lags=[10, 20],
    return_df=True
)

display(lb)

print(
    "\nLarge p-values suggest that there is not enough "
    "evidence of remaining autocorrelation."
)

def fit_and_diagnose(x, order):

    print(f"Fitting ARIMA{order}")

    fitted = ARIMA(
        x,
        order=order
    ).fit()

    display(fitted.summary())

    residual_diagnostics(fitted)

```

```
return fitted
```

Case Study 1

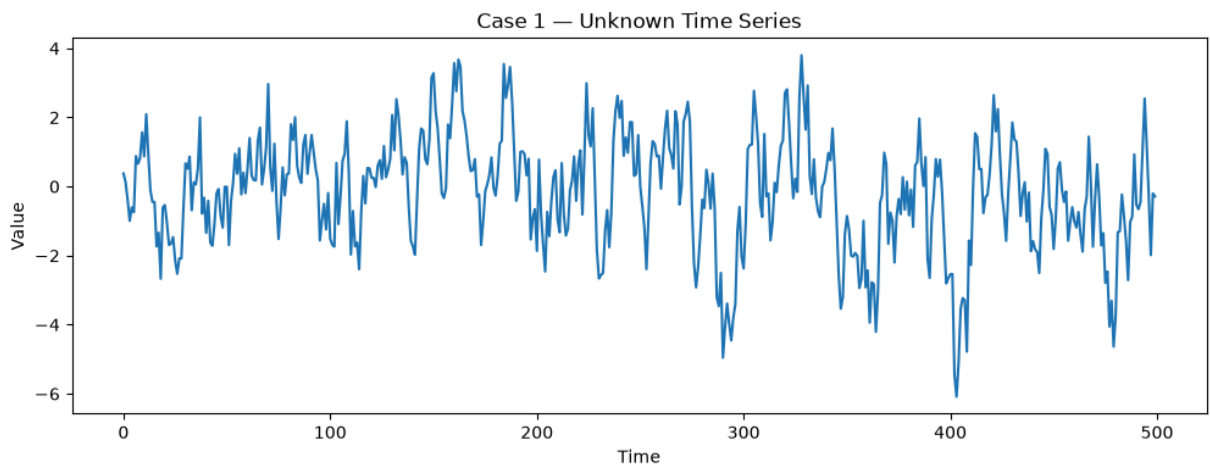
Before fitting any model:

1. Plot the series.
2. Check stationarity.
3. Inspect the ACF.
4. Inspect the PACF.
5. Write down your prediction.

Question

What model and order do you think is appropriate?

```
In [15]: plot_series(  
    x_1,  
    "Case 1 – Unknown Time Series"  
)  
  
adf_report(  
    x_1,  
    "Original Series"  
)  
  
plot_acf_pacf(  
    x_1,  
    "Case 1"  
)
```

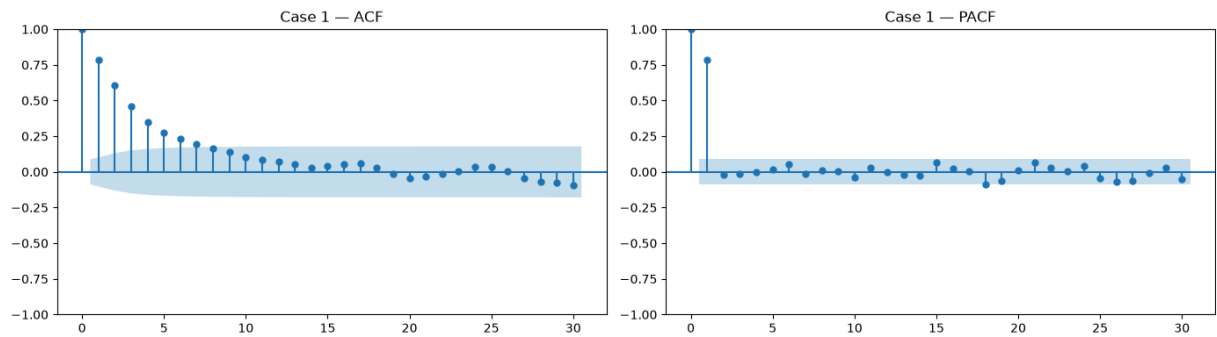


Original Series

ADF statistic: -7.7899

p-value: 0.0000

Conclusion: evidence against a unit root → stationary.



Candidate Models

Now that we have made our hypothesis, let's compare several plausible models.

We are not searching every possible ARIMA model.

Instead, we are comparing a **small set of models suggested by our investigation**. It's probably obvious that we should use AR(1) but let's see how other models compare.

```
In [16]: orders = [
    (1, 0, 0),
    (2, 0, 0),
    (0, 0, 1),
    (0, 0, 2),
    (1, 0, 1),
    (2, 0, 1)
]

results = compare_models(
    x_1,
    orders
)

display(results)

best_order = tuple(
    results.iloc[0][["p", "d", "q"]].astype(int)
)

print("Best AIC model:", best_order)
```

	Model	p	d	q	AIC	BIC
0	ARIMA(1, 0, 0)	1	0	0	1442.013240	1454.657064
1	ARIMA(2, 0, 0)	2	0	0	1443.771316	1460.629748
2	ARIMA(1, 0, 1)	1	0	1	1443.778424	1460.636857
3	ARIMA(2, 0, 1)	2	0	1	1445.727301	1466.800342
4	ARIMA(0, 0, 2)	0	0	2	1524.735717	1541.594150
5	ARIMA(0, 0, 1)	0	0	1	1625.684579	1638.328404

Best AIC model: (1, 0, 0)

Fit and Diagnose the Selected Model

```
In [52]: model_1 = fit_and_diagnose(
            x_1,
            best_order
        )
```

Fitting ARIMA(3, 0, 1)

SARIMAX Results

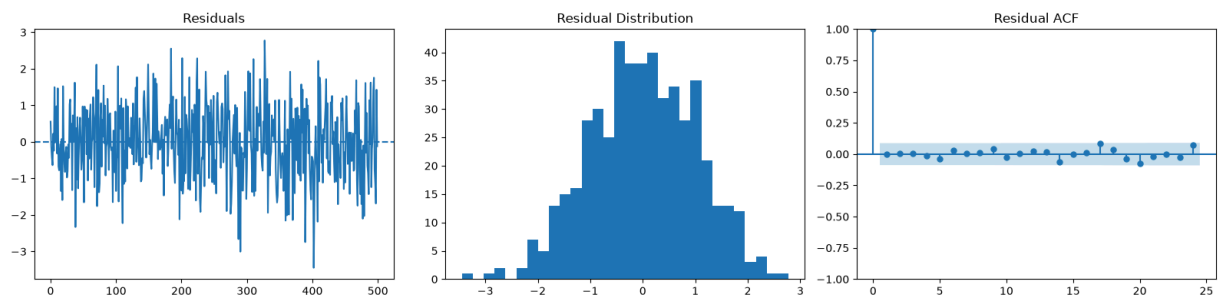
Dep. Variable:	y	No. Observations:	500
Model:	ARIMA(3, 0, 1)	Log Likelihood	-717.860
Date:	Sat, 08 Aug 2026	AIC	1447.720
Time:	03:01:21	BIC	1473.007
Sample:	0	HQIC	1457.642
	- 500		
Covariance Type:	opg		

	coef	std err	z	P> z	[0.025	0.975]
const	-0.1777	0.204	-0.871	0.384	-0.578	0.222
ar.L1	0.4930	3.963	0.124	0.901	-7.275	8.261
ar.L2	0.2294	3.150	0.073	0.942	-5.945	6.404
ar.L3	-0.0164	0.086	-0.191	0.849	-0.185	0.152
ma.L1	0.3051	3.958	0.077	0.939	-7.453	8.064
sigma2	1.0322	0.069	14.964	0.000	0.897	1.167

Ljung-Box (L1) (Q):	0.00	Jarque-Bera (JB):	1.66
Prob(Q):	0.98	Prob(JB):	0.44
Heteroskedasticity (H):	1.36	Skew:	-0.12
Prob(H) (two-sided):	0.05	Kurtosis:	2.85

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



Ljung-Box Test

	lb_stat	lb_pvalue
10	2.788353	0.985975
20	13.608304	0.849788

Large p-values suggest that there is not enough evidence of remaining autocorrelation.

Discussion

The expected pattern for AR(1) is:

- ACF gradually tails off.
- PACF has a strong spike at lag 1 and then largely cuts off.

The important point is that the ACF/PACF pattern gave us a **hypothesis**.

We then checked that hypothesis using:

- AIC/BIC
- fitted coefficients
- residual diagnostics

A model is not considered good simply because its coefficients look reasonable.

We also need the residuals to behave approximately like white noise.

Case Study 2

Use:

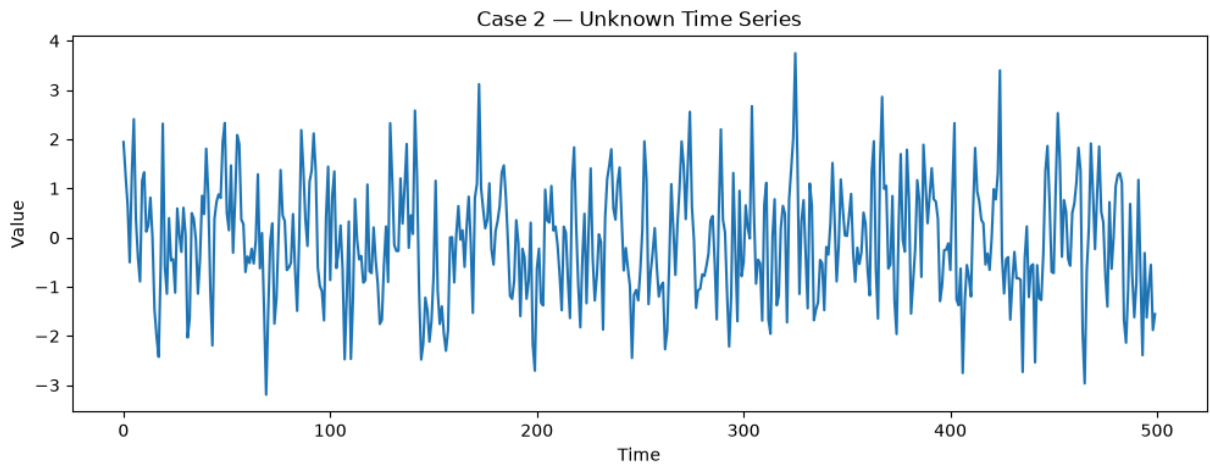
- the time-series plot
- the stationarity check
- the ACF
- the PACF

to propose a model before comparing candidates.

```
In [25]: plot_series(
          x_2,
          "Case 2 – Unknown Time Series"
        )

adf_report(
  x_2,
  "Original Series"
)
```

```
plot_acf_pacf(
    x_2,
    "Case 2"
)
```

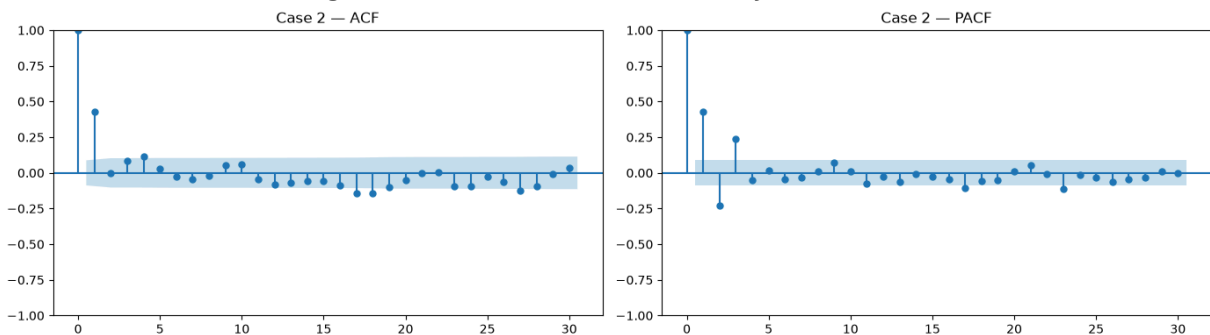


Original Series

ADF statistic: -9.7631

p-value: 0.0000

Conclusion: evidence against a unit root → stationary.



from the ACF we can suggest that we fit an MA(1) model but from the PACF we can suggest that we fit an AR(3) model, lets compare differenet variations

```
In [26]: orders = [
    (1, 0, 0),
    (2, 0, 0),
    (3, 0, 0),
    (4, 0, 0),
    (0, 0, 1),
    (0, 0, 2),
    (1, 0, 1),
    (2, 0, 1),
    (3, 0, 1),
]

results = compare_models(
    x_2,
    orders
)

display(results)
```



```
best_order = tuple(
    results.iloc[0][["p", "d", "q"]].astype(int)
)

print("Best AIC model:", best_order)
```

	Model	p	d	q	AIC	BIC
0	ARIMA(3, 0, 0)	3	0	0	1427.098213	1448.171254
1	ARIMA(3, 0, 1)	3	0	1	1427.544422	1452.832070
2	ARIMA(0, 0, 1)	0	0	1	1427.739582	1440.383407
3	ARIMA(4, 0, 0)	4	0	0	1427.829035	1453.116683
4	ARIMA(0, 0, 2)	0	0	2	1429.707145	1446.565577
5	ARIMA(1, 0, 1)	1	0	1	1429.709448	1446.567881
6	ARIMA(2, 0, 1)	2	0	1	1431.435725	1452.508765
7	ARIMA(2, 0, 0)	2	0	0	1454.141051	1470.999483
8	ARIMA(1, 0, 0)	1	0	0	1479.072887	1491.716711

Best AIC model: (3, 0, 0)

```
In [27]: model_2 = fit_and_diagnose(
    x_2,
    best_order
)
```

Fitting ARIMA(3, 0, 0)

SARIMAX Results

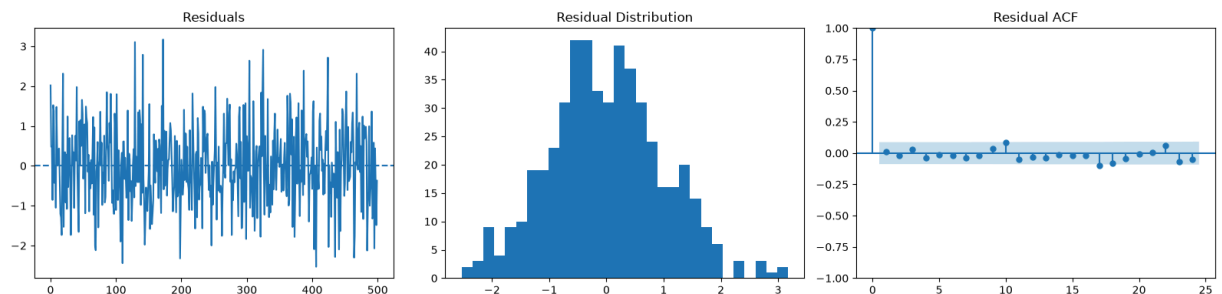
Dep. Variable:	y	No. Observations:	500
Model:	ARIMA(3, 0, 0)	Log Likelihood	-708.549
Date:	Sat, 08 Aug 2026	AIC	1427.098
Time:	02:42:23	BIC	1448.171
Sample:	0	HQIC	1435.367
	- 500		
Covariance Type:	opg		

	coef	std err	z	P> z	[0.025	0.975]
const	-0.0795	0.084	-0.942	0.346	-0.245	0.086
ar.L1	0.5847	0.045	13.077	0.000	0.497	0.672
ar.L2	-0.3549	0.054	-6.610	0.000	-0.460	-0.250
ar.L3	0.2383	0.044	5.453	0.000	0.153	0.324
sigma2	0.9953	0.063	15.701	0.000	0.871	1.120

Ljung-Box (L1) (Q):	0.06	Jarque-Bera (JB):	2.74
Prob(Q):	0.81	Prob(JB):	0.25
Heteroskedasticity (H):	0.97	Skew:	0.18
Prob(H) (two-sided):	0.86	Kurtosis:	3.04

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



Ljung-Box Test

	lb_stat	lb_pvalue
10	7.032932	0.722332
20	19.554699	0.486077

Large p-values suggest that there is not enough evidence of remaining autocorrelation.

Case Study 3

Before fitting

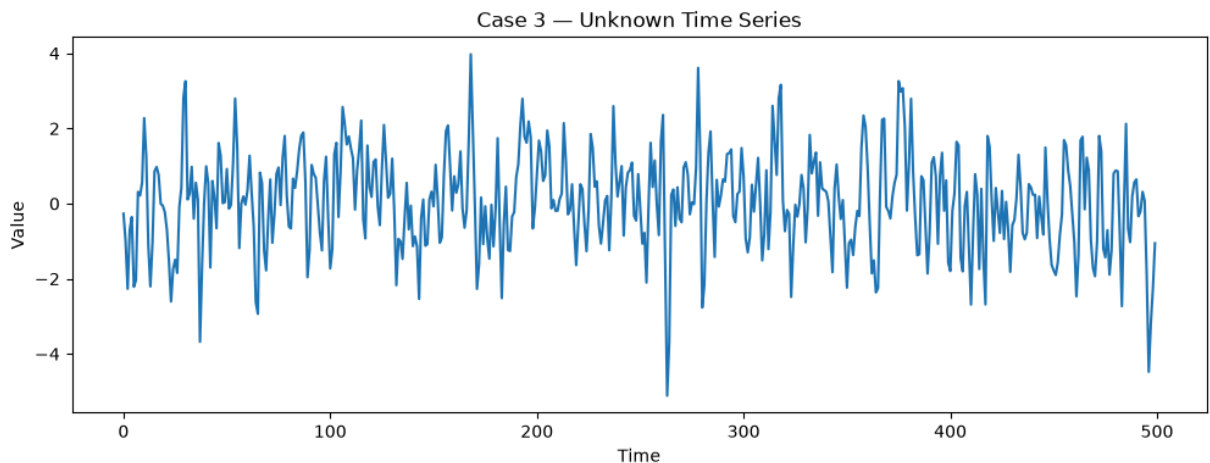
Look carefully at the difference between the ACF and PACF.

Question

Which one appears to cut off?

What model does that suggest?

```
In [29]: plot_series(  
    x_3,  
    "Case 3 – Unknown Time Series"  
)  
  
adf_report(  
    x_3,  
    "Original Series"  
)  
  
plot_acf_pacf(  
    x_3,  
    "Case 3"  
)
```

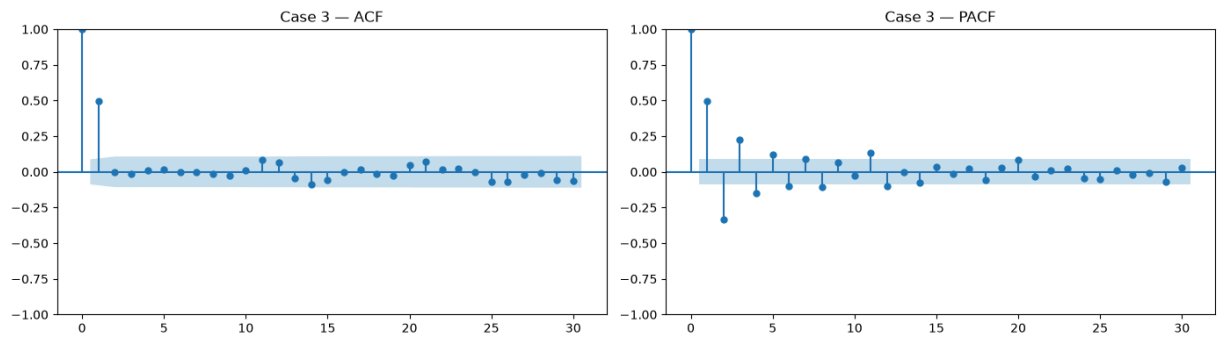


Original Series

ADF statistic: -5.7078

p-value: 0.0000

Conclusion: evidence against a unit root → stationary.



```
In [30]: orders = [
    (1, 0, 0),
    (0, 0, 1),
    (0, 0, 2),
    (1, 0, 1),
    (2, 0, 2)
]

results = compare_models(
    x_3,
    orders
)

display(results)

best_order = tuple(
    results.iloc[0][["p", "d", "q"]].astype(int)
)

print("Best AIC model:", best_order)
```

	Model	p	d	q	AIC	BIC
0	ARIMA(0, 0, 1)	0	0	1	1414.689281	1427.333105
1	ARIMA(1, 0, 1)	1	0	1	1416.666696	1433.525129
2	ARIMA(0, 0, 2)	0	0	2	1416.667488	1433.525921
3	ARIMA(2, 0, 2)	2	0	2	1420.664928	1445.952577
4	ARIMA(1, 0, 0)	1	0	0	1542.872517	1555.516342

Best AIC model: (0, 0, 1)

```
In [32]: model_3 = fit_and_diagnose(
    x_3,
    best_order
)
```

Fitting ARIMA(0, 0, 1)

SARIMAX Results

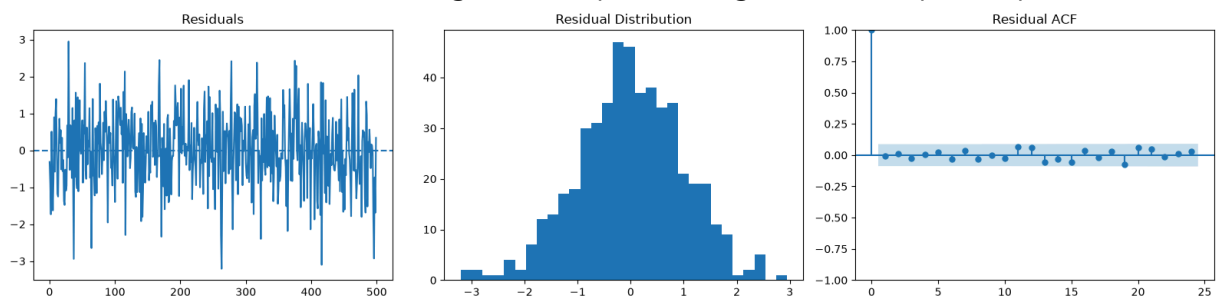
Dep. Variable:	y	No. Observations:	500
Model:	ARIMA(0, 0, 1)	Log Likelihood	-704.345
Date:	Sat, 08 Aug 2026	AIC	1414.689
Time:	02:44:09	BIC	1427.333
Sample:	0	HQIC	1419.651
	- 500		
Covariance Type:	opg		

	coef	std err	z	P> z	[0.025	0.975]
const	0.0469	0.082	0.570	0.569	-0.115	0.208
ma.L1	0.8539	0.023	37.012	0.000	0.809	0.899
sigma2	0.9771	0.060	16.419	0.000	0.860	1.094

Ljung-Box (L1) (Q):	0.01	Jarque-Bera (JB):	2.91
Prob(Q):	0.92	Prob(JB):	0.23
Heteroskedasticity (H):	1.12	Skew:	-0.16
Prob(H) (two-sided):	0.46	Kurtosis:	3.18

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



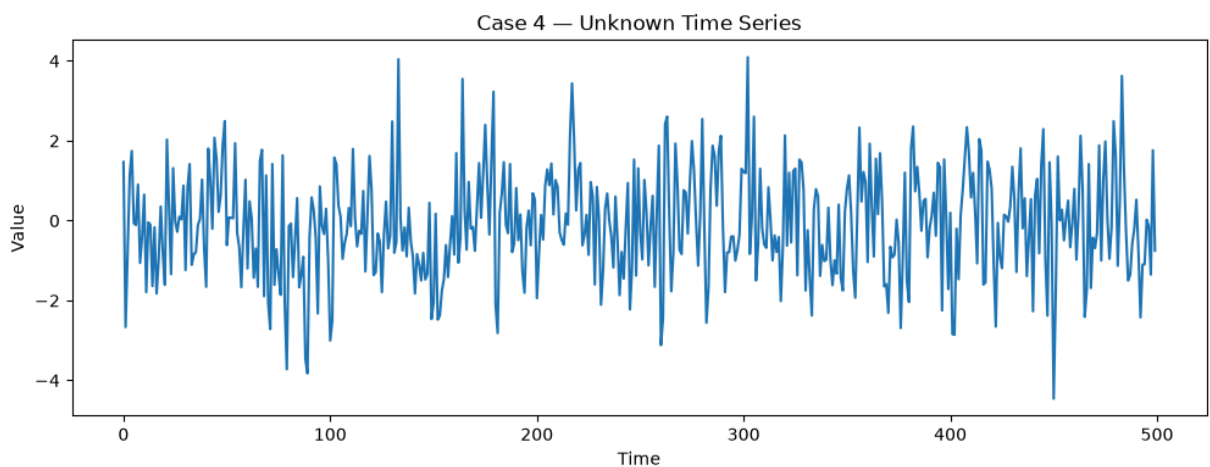
Ljung-Box Test

	lb_stat	lb_pvalue
10	2.692466	0.987765
20	16.508461	0.684622

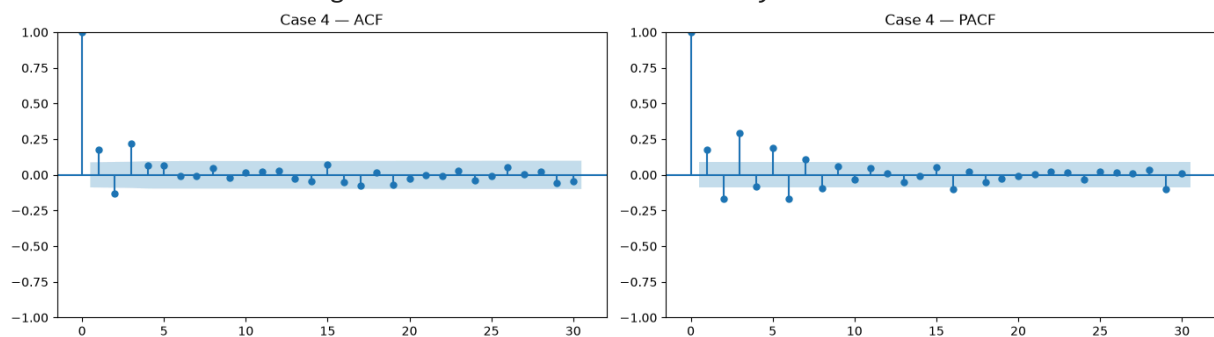
Large p-values suggest that there is not enough evidence of remaining autocorrelation.

Case Study 4

```
In [34]: plot_series(  
    x_4,  
    "Case 4 — Unknown Time Series"  
)  
  
adf_report(  
    x_4,  
    "Original Series"  
)  
  
plot_acf_pacf(  
    x_4,  
    "Case 4"  
)
```



Original Series
ADF statistic: -6.3879
p-value: 0.0000
Conclusion: evidence against a unit root → stationary.



```
In [35]: orders = [  
    (1, 0, 0),  
    (2, 0, 0),  
    (3, 0, 0),  
    (4, 0, 0),  
    (0, 0, 1),  
    (0, 0, 2),
```

```

(0, 0, 3),
(0, 0, 4),
(1, 0, 1),
(2, 0, 1),
(3, 0, 1),
(4, 0, 1),
(2, 0, 2),
(3, 0, 2)
]

results = compare_models(
    x_4,
    orders
)

display(results)

best_order = tuple(
    results.iloc[0][["p", "d", "q"]].astype(int)
)

print("Best AIC model:", best_order)

```

	Model	p	d	q	AIC	BIC
0	ARIMA(0, 0, 4)	0	0	4	1581.807106	1607.094754
1	ARIMA(3, 0, 2)	3	0	2	1583.702234	1613.204491
2	ARIMA(0, 0, 3)	0	0	3	1585.799117	1606.872158
3	ARIMA(3, 0, 1)	3	0	1	1586.259254	1611.546903
4	ARIMA(4, 0, 1)	4	0	1	1586.943734	1616.445990
5	ARIMA(2, 0, 1)	2	0	1	1593.212441	1614.285481
6	ARIMA(1, 0, 1)	1	0	1	1598.972956	1615.831389
7	ARIMA(2, 0, 2)	2	0	2	1602.078395	1627.366043
8	ARIMA(0, 0, 2)	0	0	2	1606.742467	1623.600899
9	ARIMA(4, 0, 0)	4	0	0	1633.816840	1659.104489
10	ARIMA(3, 0, 0)	3	0	0	1635.761901	1656.834941
11	ARIMA(0, 0, 1)	0	0	1	1677.854252	1690.498076
12	ARIMA(2, 0, 0)	2	0	0	1680.547557	1697.405989
13	ARIMA(1, 0, 0)	1	0	0	1693.438998	1706.082822

Best AIC model: (0, 0, 4)

```

In [37]: model_4 = fit_and_diagnose(
    x_4,
    best_order
)

```

Fitting ARIMA(0, 0, 4)

SARIMAX Results

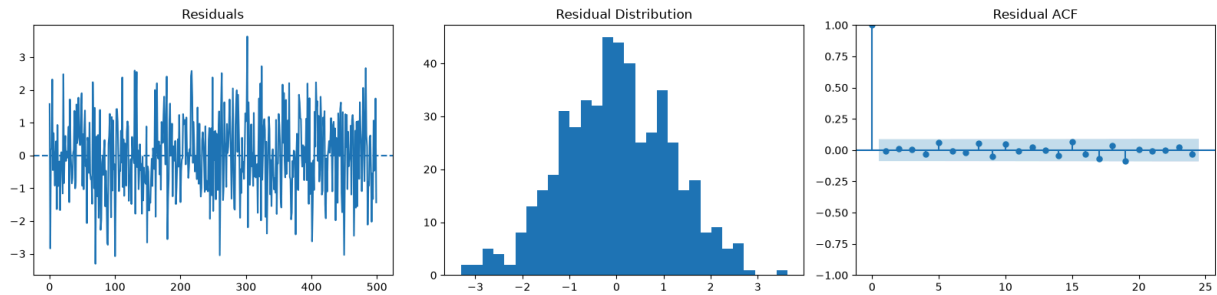
Dep. Variable:	y	No. Observations:	500
Model:	ARIMA(0, 0, 4)	Log Likelihood	-784.904
Date:	Sat, 08 Aug 2026	AIC	1581.807
Time:	02:47:47	BIC	1607.095
Sample:	0	HQIC	1591.730
	- 500		
Covariance Type:	opg		

	coef	std err	z	P> z	[0.025	0.975]
const	-0.1006	0.078	-1.294	0.196	-0.253	0.052
ma.L1	0.3918	0.043	9.099	0.000	0.307	0.476
ma.L2	-0.2905	0.048	-6.114	0.000	-0.384	-0.197
ma.L3	0.2629	0.047	5.608	0.000	0.171	0.355
ma.L4	0.1171	0.043	2.694	0.007	0.032	0.202
sigma2	1.3491	0.090	14.962	0.000	1.172	1.526

Ljung-Box (L1) (Q):	0.01	Jarque-Bera (JB):	0.68
Prob(Q):	0.94	Prob(JB):	0.71
Heteroskedasticity (H):	0.97	Skew:	-0.02
Prob(H) (two-sided):	0.86	Kurtosis:	2.82

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



Ljung-Box Test

	lb_stat	lb_pvalue
10	6.337801	0.786129
20	17.584674	0.614747

Large p-values suggest that there is not enough evidence of remaining autocorrelation.

Case Study 5

Important observation

For pure AR or pure MA models, the ACF/PACF often gives us a relatively clean cutoff.

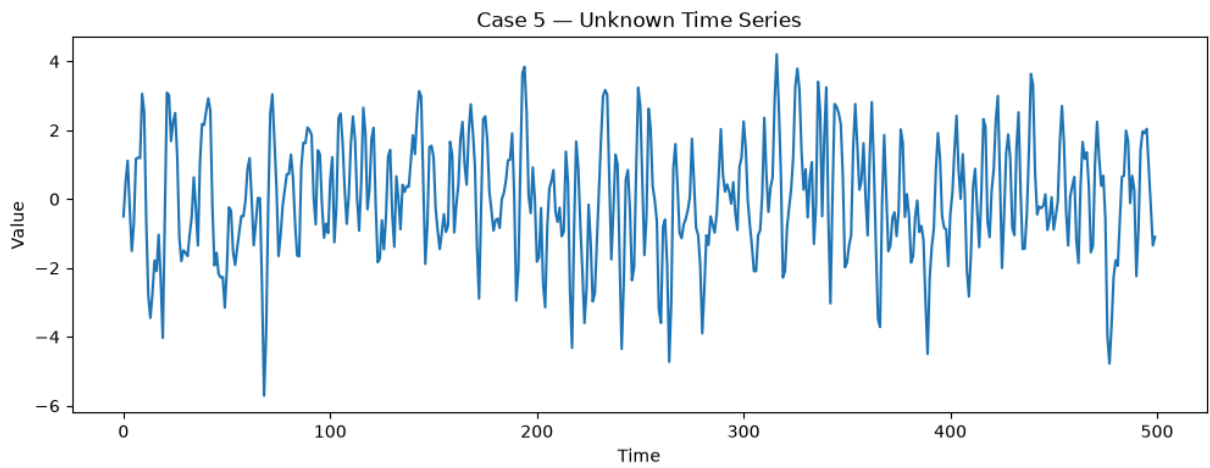
When both AR and MA components are present, the patterns become less obvious.

You should

Use ACF/PACF to generate a few plausible candidates.

Do not try to identify the exact model from a single graph.

```
In [39]: plot_series(  
          x_5,  
          "Case 5 – Unknown Time Series"  
        )  
  
adf_report(  
    x_5,  
    "Original Series"  
)  
  
plot_acf_pacf(  
    x_5,  
    "Case 5"  
)
```

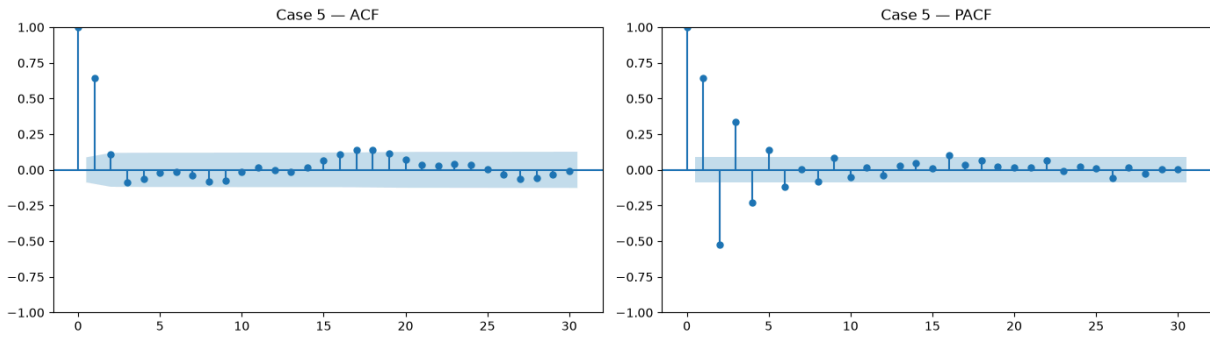


Original Series

ADF statistic: -7.1547

p-value: 0.0000

Conclusion: evidence against a unit root → stationary.



```
In [40]: orders = [
    (0, 0, 1),
    (0, 0, 2),
    (1, 0, 0),
    (2, 0, 0),
    (3, 0, 0),
    (1, 0, 1),
    (1, 0, 2),
    (2, 0, 1),
    (2, 0, 2),
    (3, 0, 1),
    (3, 0, 2)
]

results = compare_models(
    x_5,
    orders
)

display(results)

best_order = tuple(
    results.iloc[0][["p", "d", "q"]].astype(int)
)

print("Best AIC model:", best_order)
```

	Model	p	d	q	AIC	BIC
0	ARIMA(2, 0, 1)	2	0	1	1424.467975	1445.541016
1	ARIMA(3, 0, 1)	3	0	1	1426.105648	1451.393297
2	ARIMA(2, 0, 2)	2	0	2	1426.115849	1451.403497
3	ARIMA(0, 0, 2)	0	0	2	1427.530370	1444.388802
4	ARIMA(3, 0, 2)	3	0	2	1428.467447	1457.969704
5	ARIMA(1, 0, 2)	1	0	2	1428.555765	1449.628806
6	ARIMA(1, 0, 1)	1	0	1	1446.597109	1463.455541
7	ARIMA(3, 0, 0)	3	0	0	1470.532746	1491.605786
8	ARIMA(0, 0, 1)	0	0	1	1518.835549	1531.479374
9	ARIMA(2, 0, 0)	2	0	0	1528.521974	1545.380407
10	ARIMA(1, 0, 0)	1	0	0	1687.036247	1699.680071

Best AIC model: (2, 0, 1)

```
In [41]: model_5 = fit_and_diagnose(
            x_5,
            best_order
        )
```

Fitting ARIMA(2, 0, 1)

SARIMAX Results

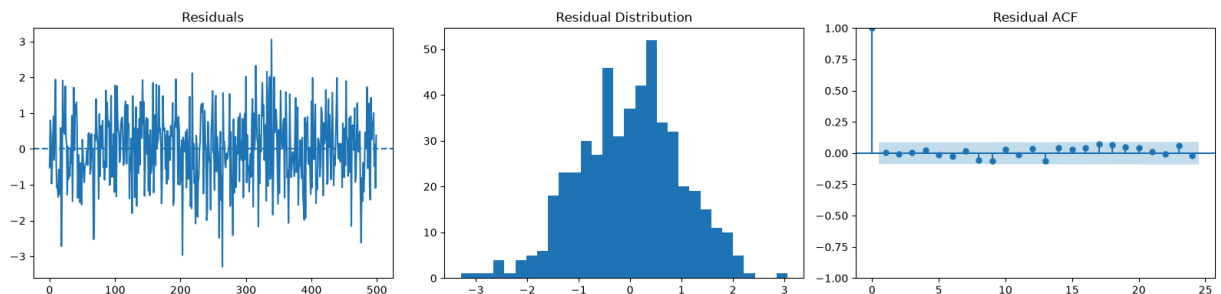
Dep. Variable:	y	No. Observations:	500
Model:	ARIMA(2, 0, 1)	Log Likelihood	-707.234
Date:	Sat, 08 Aug 2026	AIC	1424.468
Time:	02:50:58	BIC	1445.541
Sample:	0	HQIC	1432.737
	- 500		
Covariance Type:	opg		

	coef	std err	z	P> z	[0.025	0.975]
const	0.0211	0.112	0.188	0.851	-0.199	0.242
ar.L1	0.5568	0.053	10.600	0.000	0.454	0.660
ar.L2	-0.2568	0.054	-4.758	0.000	-0.363	-0.151
ma.L1	0.7320	0.041	17.842	0.000	0.652	0.812
sigma2	0.9871	0.064	15.521	0.000	0.862	1.112

Ljung-Box (L1) (Q):	0.03	Jarque-Bera (JB):	2.02
Prob(Q):	0.87	Prob(JB):	0.36
Heteroskedasticity (H):	1.07	Skew:	-0.16
Prob(H) (two-sided):	0.66	Kurtosis:	2.98

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



Ljung-Box Test

	lb_stat	lb_pvalue
10	4.987181	0.892033
20	16.844997	0.663015

Large p-values suggest that there is not enough evidence of remaining autocorrelation.

Case Study 6

Important question

Should we inspect the ACF/PACF of the original series and immediately decide p and q ?

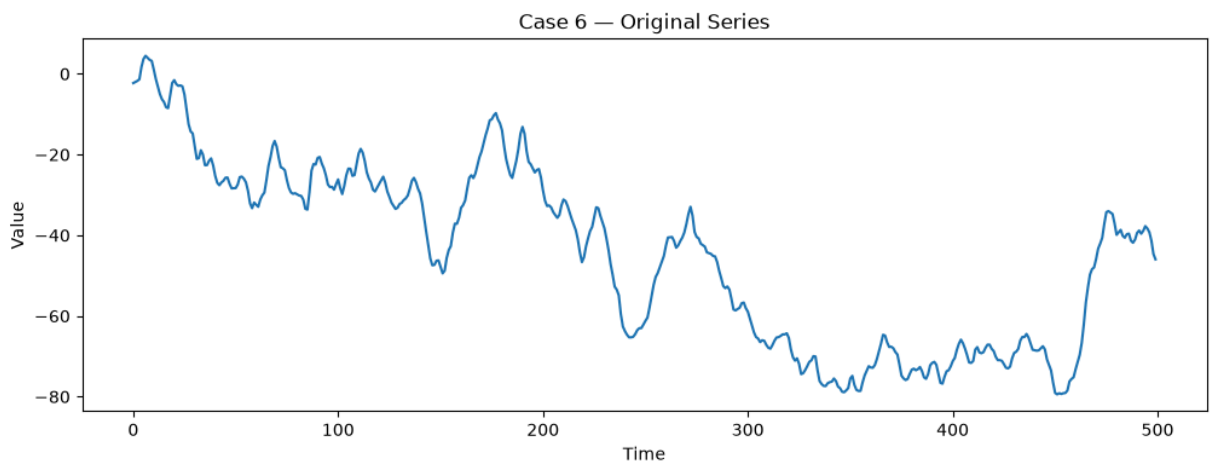
No.

First we need to determine whether differencing is required.

Workflow

Original series → **stationarity** → **difference** → **stationarity again** → **ACF/PACF of differenced series**

```
In [43]: plot_series(  
          x_6,  
          "Case 6 - Original Series"  
        )  
  
adf_report(  
    x_6,  
    "Original Series"  
)
```



Original Series

ADF statistic: -2.3833

p-value: 0.1465

Conclusion: insufficient evidence to reject the unit-root null → non-stationary.

```
In [44]: x_6_diff = x_6.diff().dropna()  
  
plot_series(  
    x_6_diff,
```

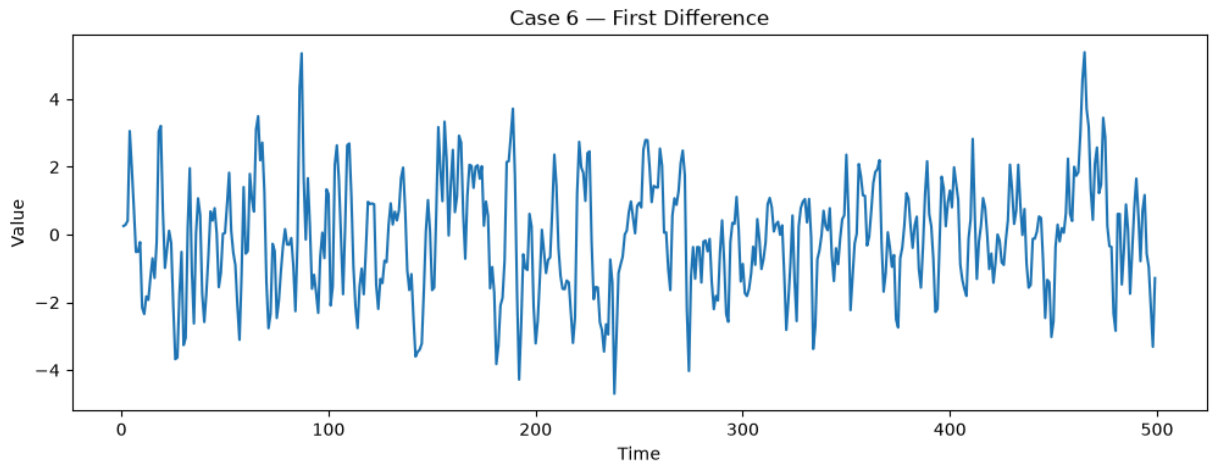
```

    "Case 6 – First Difference"
)

adf_report(
  x_6_diff,
  "First Difference"
)

plot_acf_pacf(
  x_6_diff,
  "Case 6 – Differenced Series"
)

```

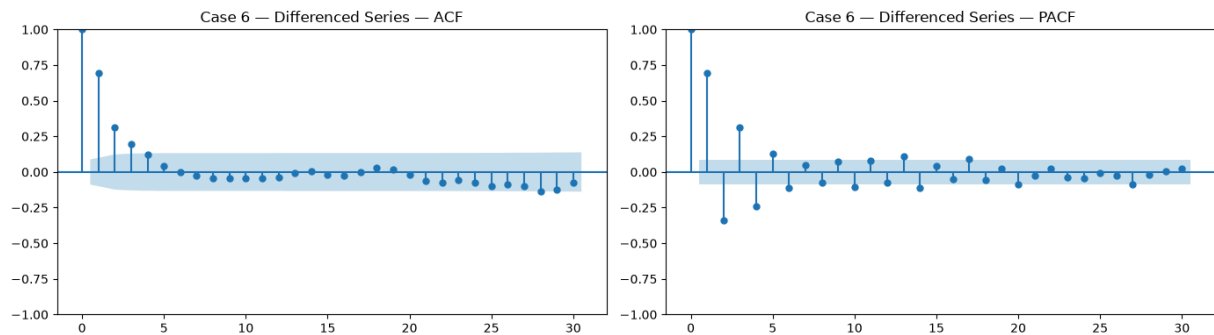


First Difference

ADF statistic: -4.9327

p-value: 0.0000

Conclusion: evidence against a unit root → stationary.



Candidate Models

Based on the differenced series, propose a few values for p and q .

Remember:

We are now modelling:

$$\Delta X_t$$

not the original X_t .

Therefore all candidate models should have:

$$d = 1$$

```
In [45]: orders = [
    (1, 1, 0),
    (0, 1, 1),
    (1, 1, 1),
    (1, 1, 2),
    (2, 1, 1),
    (2, 1, 2),
    (2, 1, 3),
    (3, 1, 2)
]

results = compare_models(
    x_6,
    orders
)

display(results)

best_order = tuple(
    results.iloc[0][["p", "d", "q"]].astype(int)
)

print("Best AIC model:", best_order)
```

	Model	p	d	q	AIC	BIC
0	ARIMA(1, 1, 2)	1	1	2	1407.136398	1423.986823
1	ARIMA(2, 1, 1)	2	1	1	1407.727685	1424.578110
2	ARIMA(2, 1, 2)	2	1	2	1409.094470	1430.157501
3	ARIMA(1, 1, 1)	1	1	1	1409.194067	1421.831886
4	ARIMA(3, 1, 2)	3	1	2	1411.062759	1436.338395
5	ARIMA(2, 1, 3)	2	1	3	1411.133856	1436.409493
6	ARIMA(0, 1, 1)	0	1	1	1465.468501	1473.893713
7	ARIMA(1, 1, 0)	1	1	0	1592.954696	1601.379908

Best AIC model: (1, 1, 2)

```
In [46]: model_6 = fit_and_diagnose(
    x_6,
    best_order
)
```

Fitting ARIMA(1, 1, 2)

SARIMAX Results

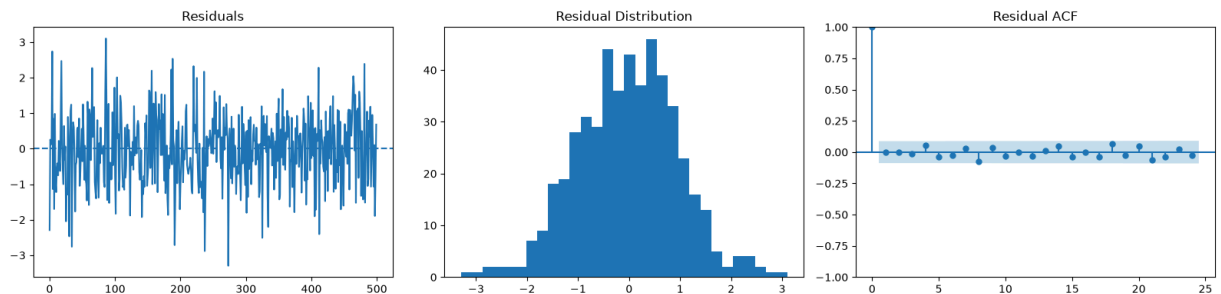
Dep. Variable:	y	No. Observations:	500
Model:	ARIMA(1, 1, 2)	Log Likelihood	-699.568
Date:	Sat, 08 Aug 2026	AIC	1407.136
Time:	02:55:28	BIC	1423.987
Sample:	0	HQIC	1413.749
	- 500		
Covariance Type:	opg		

	coef	std err	z	P> z	[0.025	0.975]
ar.L1	0.5816	0.092	6.293	0.000	0.400	0.763
ma.L1	0.6598	0.109	6.049	0.000	0.446	0.874
ma.L2	-0.2620	0.107	-2.442	0.015	-0.472	-0.052
sigma2	0.9614	0.059	16.250	0.000	0.845	1.077

Ljung-Box (L1) (Q):	0.00	Jarque-Bera (JB):	0.75
Prob(Q):	0.98	Prob(JB):	0.69
Heteroskedasticity (H):	0.85	Skew:	-0.05
Prob(H) (two-sided):	0.30	Kurtosis:	3.16

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



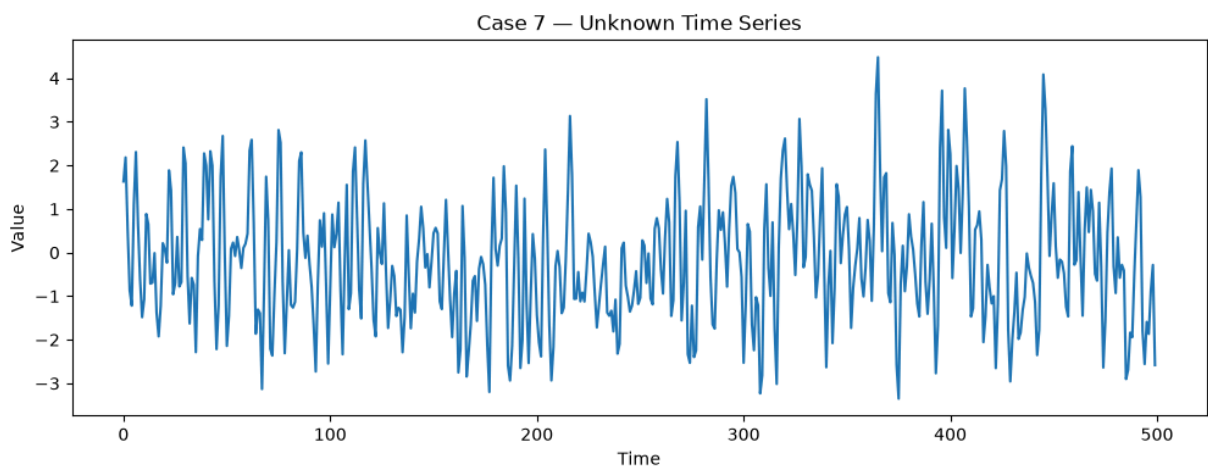
Ljung-Box Test

	lb_stat	lb_pvalue
10	7.165102	0.709772
20	14.334846	0.813121

Large p-values suggest that there is not enough evidence of remaining autocorrelation.

Case Study 7

```
In [48]: plot_series(  
    x_7,  
    "Case 7 — Unknown Time Series"  
)  
  
adf_report(  
    x_7,  
    "Original Series"  
)  
  
plot_acf_pacf(  
    x_7,  
    "Case 7"  
)
```

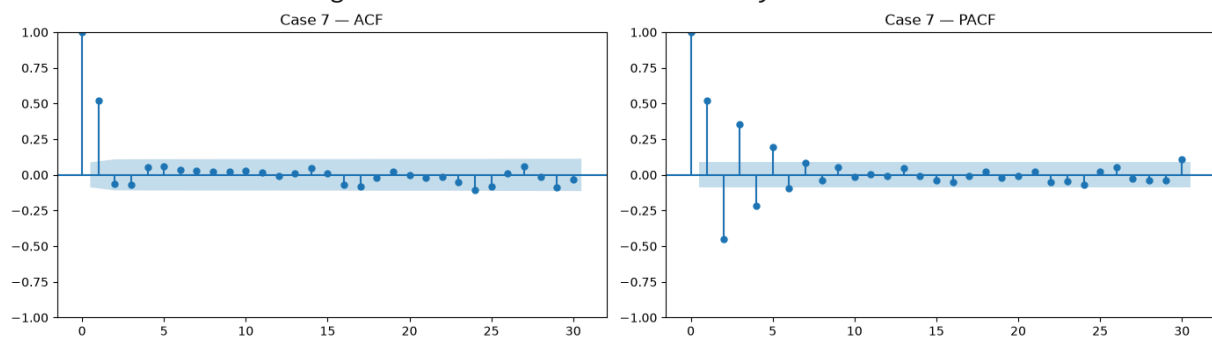


Original Series

ADF statistic: -6.9772

p-value: 0.0000

Conclusion: evidence against a unit root → stationary.



```
In [50]: orders = [  
    (0, 0, 1),  
    (0, 0, 2),  
    (1, 0, 0),  
    (2, 0, 0),  
    (3, 0, 0),  
    (1, 0, 1),
```

```

(1, 0, 2),
(2, 0, 1),
(2, 0, 2),
(3, 0, 1),
(3, 0, 2),
(3, 0, 3),
(4, 0, 2),
(5, 0, 1),
(5, 0, 2)
]

results = compare_models(
    x_7,
    orders
)

display(results)

best_order = tuple(
    results.iloc[0][["p", "d", "q"]].astype(int)
)

print("Best AIC model:", best_order)

```

	Model	p	d	q	AIC	BIC
0	ARIMA(3, 0, 1)	3	0	1	1385.098655	1410.386303
1	ARIMA(2, 0, 2)	2	0	2	1385.289355	1410.577004
2	ARIMA(3, 0, 2)	3	0	2	1387.024951	1416.527208
3	ARIMA(2, 0, 1)	2	0	1	1387.471412	1408.544453
4	ARIMA(3, 0, 3)	3	0	3	1388.158692	1421.875557
5	ARIMA(5, 0, 1)	5	0	1	1388.439503	1422.156367
6	ARIMA(4, 0, 2)	4	0	2	1388.595189	1422.312053
7	ARIMA(1, 0, 2)	1	0	2	1389.938584	1411.011625
8	ARIMA(5, 0, 2)	5	0	2	1391.032442	1428.963915
9	ARIMA(0, 0, 2)	0	0	2	1396.486572	1413.345004
10	ARIMA(1, 0, 1)	1	0	1	1408.732433	1425.590866
11	ARIMA(0, 0, 1)	0	0	1	1423.717124	1436.360948
12	ARIMA(3, 0, 0)	3	0	0	1448.016582	1469.089622
13	ARIMA(2, 0, 0)	2	0	0	1519.162130	1536.020563
14	ARIMA(1, 0, 0)	1	0	0	1632.277326	1644.921150

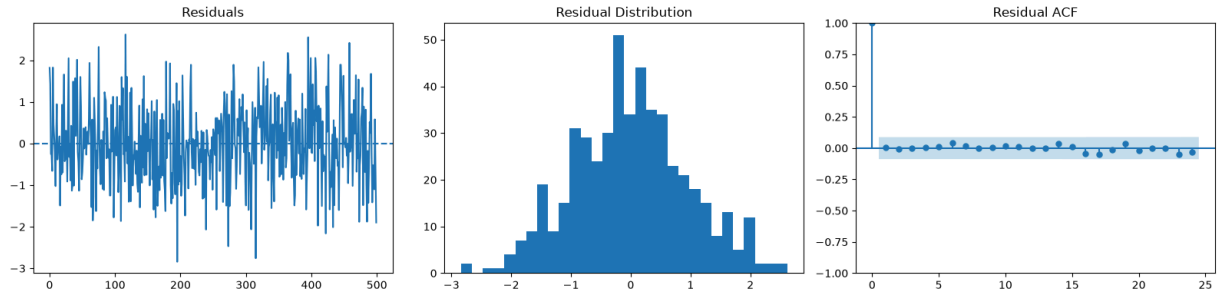
Best AIC model: (3, 0, 1)

```
In [51]: model_7 = fit_and_diagnose(  
        x_7,  
        best_order  
    )
```

Fitting ARIMA(3, 0, 1)
SARIMAX Results

Dep. Variable:		y	No. Observations:		500	
Model:		ARIMA(3, 0, 1)		Log Likelihood		-686.549
Date:		Sat, 08 Aug 2026		AIC		1385.099
Time:		02:57:41		BIC		1410.386
Sample:		0		HQIC		1395.021
		- 500				
Covariance Type:		opg				
	coef	std err	z	P> z	[0.025	0.975]
const	-0.1884	0.091	-2.080	0.038	-0.366	-0.011
ar.L1	0.3608	0.063	5.703	0.000	0.237	0.485
ar.L2	-0.3097	0.060	-5.125	0.000	-0.428	-0.191
ar.L3	0.1158	0.053	2.196	0.028	0.012	0.219
ma.L1	0.7497	0.045	16.739	0.000	0.662	0.837
sigma2	0.9089	0.060	15.102	0.000	0.791	1.027
Ljung-Box (L1) (Q):		0.00	Jarque-Bera (JB):		0.69	
Prob(Q):		0.96	Prob(JB):		0.71	
Heteroskedasticity (H):		1.15	Skew:		0.07	
Prob(H) (two-sided):		0.36	Kurtosis:		2.89	

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).



Ljung-Box Test

	lb_stat	lb_pvalue
10	1.284772	0.999464
20	5.279280	0.999578

Large p-values suggest that there is not enough evidence of remaining autocorrelation.

Your task

For dataset D:

1. Plot the series.
2. Check stationarity.
3. Determine whether differencing is needed.
4. Inspect ACF and PACF.
5. Propose 2–4 candidate models.
6. Compare the candidates.
7. Fit the selected model.
8. Diagnose the residuals.
9. Decide whether the model is adequate.

In []: